Table 6. Random parameters binary logit model with heterogeneity in means and variances of crash injury severity (serious injury/fatal vs. PDO/minor injury), single-vehicle crashes without a motorcycle, Thailand, 2016–2023 (yearly periods), adaptive fused temporal structure.

Variable description	Estimated parameter	*t*-statistic	Marginal effects	
			PDO / minor injury	Serious injury / fatal
Constant [serious injury / fatal]				
Constant [2016]	–2.089	–25.72		
Constant [2017–2019]	–2.270	–30.43		
Constant [2020]	–2.321	–28.39		
Constant [2021]	–2.457	–29.63		
Constant [2022–2023]	–2.577	–32.55		
Variables producing the same parameter value across all periods [2016–2023]				
Vehicle involvement				
One passenger car involved (1 if yes, 0 otherwise) [2016–2023]	0.157	4.82	–0.0121	0.0121
One 6-wheel truck involved (1 if yes, 0 otherwise) [2016–2023]	–0.385	–5.30	0.0261	–0.0261
One 10-wheel truck involved (1 if yes, 0 otherwise) [2016–2023]	–0.743	–8.29	0.0456	–0.0456
One truck with more than 10 wheels involved (1 if yes, 0 otherwise) [2016–2023]	–0.909	–13.70	0.0548	–0.0548
Van involved (1 if yes, 0 otherwise) [2016–2023]	0.728	8.21	–0.0671	0.0671
Crash type				
Rear-end collision in the same lane (1 if yes, 0 otherwise) [2016–2023]	1.020	8.26	–0.1019	0.1019
Vehicle mounted a traffic island (1 if yes, 0 otherwise) [2016–2023]	–0.636	–9.59	0.0428	–0.0428
Ran off road on a left bend and hit an object (1 if yes, 0 otherwise) [2016–2023]	–5.803	–3.15	0.0202	–0.0202
(Standard deviation of parameter distribution; 17.3% above zero)	6.158	4.04		
Ran off road on a left bend (1 if yes, 0 otherwise) [2016–2023]	0.303	2.23	–0.0249	0.0249
Ran off road on a right bend (1 if yes, 0 otherwise) [2016–2023]	0.378	2.75	–0.0317	0.0317
Crash cause				
Speeding (1 if yes, 0 otherwise) [2016–2023]	0.468	8.28	–0.0328	0.0328
Cutting in too closely (1 if yes, 0 otherwise) [2016–2023]	0.388	3.10	–0.0327	0.0327
Roadway characteristics				
Crash on the main carriageway (1 if yes, 0 otherwise) [2016–2023]	–0.405	–5.45	0.0274	–0.0274
Road under construction (1 if yes, 0 otherwise) [2016–2023]	0.171	1.73	–0.0135	0.0135
Painted median (1 if yes, 0 otherwise) [2016–2023]	–0.218	–3.37	0.0155	–0.0155
Raised or earth median (1 if yes, 0 otherwise) [2016–2023]	–0.190	–4.65	0.0139	–0.0139
Horizontal curve (1 if yes, 0 otherwise) [2016–2023]	–0.983	–3.73	–0.0084	0.0084
(Standard deviation of parameter distribution; 31.9% above zero)	2.087	6.77		
Concrete pavement surface (1 if yes, 0 otherwise) [2016–2023]	–0.592	–2.98	0.0058	–0.0058
(Standard deviation of parameter distribution; 33.3% above zero)	1.371	5.34		
T-intersection (1 if yes, 0 otherwise) [2016–2023]	–0.364	–3.58	0.0247	–0.0247
Bridge section (1 if yes, 0 otherwise) [2016–2023]	0.493	4.88	–0.0427	0.0427
Environmental characteristics				
Wet road surface (1 if yes, 0 otherwise) [2016–2023]	–0.253	–5.86	0.0182	–0.0182
Temporal characteristics				
Weekend (1 if yes, 0 otherwise) [2016–2023]	0.064	2.04	–0.0048	0.0048
Songkran festival period (1 if yes, 0 otherwise) [2016–2023]	–0.841	–3.54	–0.0124	0.0124
(Standard deviation of parameter distribution; 34.5% above zero)	2.101	6.47		
New Year festival period (1 if yes, 0 otherwise) [2016–2023]	–0.877	–3.69	–0.0148	0.0148
(Standard deviation of parameter distribution; 34.5% above zero)	2.195	6.80		
Public holiday (1 if yes, 0 otherwise) [2016–2023]	–0.167	–1.69	0.0120	–0.0120
Midnight / early morning (1 if yes, 0 otherwise) [2016–2023]	0.090	2.80	–0.0069	0.0069
Region				
Southern region (1 if yes, 0 otherwise) [2016–2023]	0.155	4.06	–0.0120	0.0120
Variables with temporally varying parameters				
Vehicle involvement				
Pedestrian involved (1 if yes, 0 otherwise) [2016–2022]	6.191	7.36	–0.6675	0.6675
(Standard deviation of parameter distribution; 95.3% above zero)	3.694	3.65		

Variable description	Estimated parameter	*t*-statistic	Marginal effects	
			PDO / minor injury	Serious injury / fatal
Pedestrian involved (1 if yes, 0 otherwise) [2023]	5.851	13.67	–0.7411	0.7411
Crash type				
Ran off road to the left and hit an object (1 if yes, 0 otherwise) [2016–2018]	–0.463	–5.97	0.0376	–0.0376
Ran off road to the left and hit an object (1 if yes, 0 otherwise) [2019–2023]	–0.218	–3.33	0.0147	–0.0147
Ran off road to the right and hit an object (1 if yes, 0 otherwise) [2016–2022]	–0.369	–5.83	0.0271	–0.0271
Ran off road to the right and hit an object (1 if yes, 0 otherwise) [2023]	–0.465	–4.17	0.0262	–0.0262
Ran off road to the left (1 if yes, 0 otherwise) [2016–2021]	0.568	8.22	–0.0524	0.0524
Ran off road to the left (1 if yes, 0 otherwise) [2022–2023]	0.870	8.30	–0.0677	0.0677
Ran off road on a right bend and hit an object (1 if yes, 0 otherwise) [2016–2018]	–0.642	–4.27	0.0478	–0.0478
Ran off road on a right bend and hit an object (1 if yes, 0 otherwise) [2019–2020]	–0.295	–1.90	0.0219	–0.0219
Ran off road on a right bend and hit an object (1 if yes, 0 otherwise) [2021–2023]	–0.601	–3.73	0.0320	–0.0320
Ran off road to the right (1 if yes, 0 otherwise) [2016–2017]	0.817	8.18	–0.0895	0.0895
Ran off road to the right (1 if yes, 0 otherwise) [2018–2023]	0.458	5.68	–0.0369	0.0369
Crash cause				
Driver fell asleep (1 if yes, 0 otherwise) [2016–2020]	0.940	12.92	–0.0969	0.0969
Driver fell asleep (1 if yes, 0 otherwise) [2021–2023]	0.619	7.01	–0.0455	0.0455
Roadway characteristics				
Barrier median (1 if yes, 0 otherwise) [2016–2018]	–0.673	–7.06	0.0503	–0.0503
Barrier median (1 if yes, 0 otherwise) [2019–2023]	–1.667	–4.38	0.0595	–0.0595
(Standard deviation of parameter distribution; 10.5% above zero)	1.327	3.06		
Environmental characteristics				
Dark, no street lighting (1 if yes, 0 otherwise) [2016–2022]	0.557	11.33	–0.0497	0.0497
Dark, no street lighting (1 if yes, 0 otherwise) [2023]	0.315	2.42	–0.0208	0.0208
Region				
Northern region (1 if yes, 0 otherwise) [2019–2023]	–1.083	–3.72	0.0109	–0.0109
(Standard deviation of parameter distribution; 28.1% above zero)	1.863	5.52		
Western region (1 if yes, 0 otherwise) [2018–2023]	0.444	8.24	–0.0354	0.0354
Heterogeneity in the means of random parameters				
Pedestrian involved [2016–2022]: Northeastern region [2016–2023]	2.883	3.24		
Horizontal curve [2016–2023]: Dark, with street lighting [2016–2023]	0.444	4.40		
Horizontal curve [2016–2023]: Raining [2020–2023]	–0.882	–4.89		
Heterogeneity in the variances of random parameters				
Horizontal curve [2016–2023]: Raining [2016–2019]	–0.433	–3.42		
Model statistics				
Number of observations		73,731		
Number of estimated parameters		65		
Log-likelihood at zero, LL(0)		–51106.43		
Log-likelihood with constant only, LL(C)		–26810.27		
Log-likelihood at convergence, LL(β)		–23646.87		
McFadden pseudo $\rho^2 = 1 - LL(\beta)/LL(0)$		0.537		
AIC / BIC		47,423.7 / 48,022.3		
Random parameter distribution; simulation		Normal; 2,000 Halton draws		

Notes. A positive parameter increases the probability of a serious injury/fatal outcome. Reference categories: Central region, daytime, no median, one pickup truck. Periods in brackets share one parameter value; for a variable with temporally varying parameters, periods not shown have a parameter of zero. Temporal structure selected by the adaptive fused lasso in the fixed-parameter binary logit (Stage 1): the sparsest penalty whose structure is not rejected at the 90% level against the separate-period logits ($LR < \chi^2_{0,90}(df)$). The stage at which the structure was chosen (fixed-parameter or random parameters model) is the one with the lower BIC among models not rejected at the 90% level against the fully unconstrained models (Table 3; all candidates in Table D1). Random parameters (at most eight), heterogeneity in means (at most three terms) and in variances (at most one term) were searched with 100 Halton draws at the 90% level and confirmed with 2,000 draws; non-significant parameters were then removed, so all parameters except constants and the means of random parameters are statistically significant at the 90% level (a parameter is kept random only while its standard deviation is significant). A variable used in a heterogeneity function enters only the random parameter shown, for crashes in the shared periods, and is removed from the main utility. Marginal effects are average changes in the simulated unconditional probability (2,000 Halton draws) when the indicator switches from 0 to 1, averaged over crashes in the parameter's periods; for random parameters they include the heterogeneity terms. PDO/minor = –(serious/fatal). Percentage above zero evaluated at the heterogeneity variables equal to zero.